

# Clinical and Neurophysiological Effects of Robotically Delivered fMRI-Guided Personalized Transcranial Magnetic Stimulation Therapy for Depression

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## Keywords

Depression · Personalization · Precision medicine · Transcranial magnetic stimulation · fMRI

## Abstract

**Introduction:** Repetitive transcranial magnetic stimulation (rTMS) of the left dorsolateral prefrontal cortex (DLPFC) is an established treatment for refractory major depressive disorder (MDD), but treatment outcomes vary substantially from person to person. Recent evidence suggests that incorporating neuroimaging-based targeting may help improve clinical outcomes. Here, we report the initial clinical outcomes of our open-label fMRI-personalized treatment protocol from the Queensland Neurostimulation Centre. **Methods:** This open-label, nonrandomized study was conducted between November 2021 and September 2024. Participants were a referred sample aged between 19 and 84, meeting the criteria for treatment-resistant MDD ( $N = 61$ ).

They received 20- or 30-weekday sessions of DLPFC rTMS. The stimulation site was personalized using each individual's fMRI brain connectivity data. **Results:** The primary outcome was change in the Montgomery-Åsberg Depression Rating Scale (MADRS). MADRS was lower post-treatment ( $d = 1.78$ ,  $p < 0.001$ ), with 52% and 33% response and remission rates observed. Likewise, anxiety scores (Hamilton Anxiety Rating Scale) were lower post-treatment ( $d = 1.27$ ,  $p < 0.001$ ), with 46% and 28% response and remission rates observed. The treatment was most effective in patients who qualified for randomized controlled trials (RCTs;  $N = 19$ , MADRS response = 74%, remission = 53%) and least effective in patients with bipolar or neurological disorders ( $N = 8$ , MADRS response = 38%, remission = 25%). Neurophysiologically, functional brain connectivity in the personalized DLPFC-subgenual cingulate cortex pathway was less anticorrelated post-treatment ( $d = 0.63$ ,  $p < 0.001$ ). **Conclusion:** Our findings provide new clinical and neurophysiological evidence supporting the high effectiveness of fMRI connectivity-guided personalized rTMS for

MDD, especially in individuals without complex comorbidities. The results encourage future RCTs to assess the superiority of personalized targeting over standard TMS.

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## Introduction

Repetitive transcranial magnetic stimulation (rTMS) of the left DLPFC is an established treatment for refractory major depressive disorder (MDD) that has become increasingly available and affordable. While approximately 45% of individuals meet response criteria [1], clinical outcomes to rTMS are variable.

Variability in rTMS antidepressant outcomes may be partly attributed to the standard “one-size-fits-all” targeting of the DLPFC [2]. Several lines of evidence suggest that clinical outcomes are better when rTMS is applied to DLPFC sites functionally connected to the subgenual cingulate cortex (SGC) [2–6]. To translate these findings to the clinic, we developed a computational method for personalizing rTMS stimulation targets based on an individual’s unique spatial pattern of SGC-DLPFC resting-state functional connectivity (RSFC) [7].

Here, we report clinical outcomes of our personalized protocol in 61 participants with MDD treated at the Queensland Neurostimulation Centre (QNC), benchmarking them against the 31% response and 23% remission rate reported in open-label studies of DLPFC rTMS [8]. In addition, we utilized a pre-established consensus classification framework [9] to examine the effect of clinical presentation on treatment outcomes. Participants were assigned to three groups: (G1) MDD eligible for randomized clinical trials ( $N = 19$ ), (G2) MDD with complex psychiatric comorbidities ( $N = 34$ ), and (G3) bipolar depression or structural neurological disorders ( $N = 8$ ). Finally, to assess whether our treatment approach modulated the targeted DLPFC-SGC RSFC pathway, we measured changes in RSFC from pre- to post-treatment in a subset of participants ( $N = 28$ ).

## Methods

Data were collected as part of QNC, a neuro-modulation clinic ( $N = 69$ ). Psychiatrists and general practitioners referred patients who had either a primary or secondary diagnosis of MDD. During the initial intake session, each referral was screened by a QNC psychiatrist (B.B.). All individuals met the criteria for treatment-

resistant depression, having undertaken two or more trials of pharmacological treatments and, if appropriate, a course of psychotherapy with persisting symptoms (see online suppl. Tables 1–3 for details regarding prior and current medication use; for all online suppl. material, see <https://doi.org/10.1159/000545692>). Treatment procedures were approved by the Australian TGA (DAP21-0020539). The research protocols were approved by Local Ethics Committees (UnitingCare and QIMRB), and participants provided written informed consent. Eight participants did not complete the full rTMS treatment (three withdrew due to side effects, three discontinued due to other reasons, and two consented to treatment but did not commence), leaving a final sample of 61 (Table 1).

As mentioned above, to examine the effect of clinical presentations, we a priori employed a previously published Delphi method-based consensus grouping approach [9]. Group 1 (G1) included patients who would qualify for RCTs based on the consensus definition of treatment-resistant depression, whereby treatment resistance cannot be in part explained by the presence of complex psychiatric or neurological comorbidity or better explained by another primary disorder [9]. Group 2 (G2) included patients who did not fit the RCT criteria delineated in Sforzini et al. [9] due to psychiatric comorbidities (e.g., comorbid obsessive compulsive disorder). Group 3 (G3) included patients who did not fit the RCT criteria due to bipolar depression or structural neurological disorders. While the sample size for G3 is small, their outcomes are clinically useful given the population is largely underreported in the literature.

Following the initial referral and recruitment, an MRI scan (3T) of brain anatomy and a 15-min eyes-open resting-state functional scan were acquired for each individual. Images were preprocessed using a standard pipeline (online suppl. Material) and each participant’s rTMS target was defined using our personalization method [7]. In brief, the right SGC signal-to-noise was increased by weighting each individual’s data by a group-averaged connectivity map derived from the Human Connectome Project. The weighted SGC time series was then correlated with the left DLPFC and the largest anticorrelated cluster was chosen as the personalized stimulation target. Clustering was performed across various thresholds to identify a reliable coordinate. In some cases, multiple coordinates were identified, and the final stimulation site was based on tolerability (stimulation of ventral-anterior clusters is less tolerable compared to dorso-posterior clusters) and other factors (e.g., proximity of the local maxima to gyri).

rTMS was precisely and consistently administered to the personalized left DLPFC site using a TMS robot

**Table 1.** Population characteristics

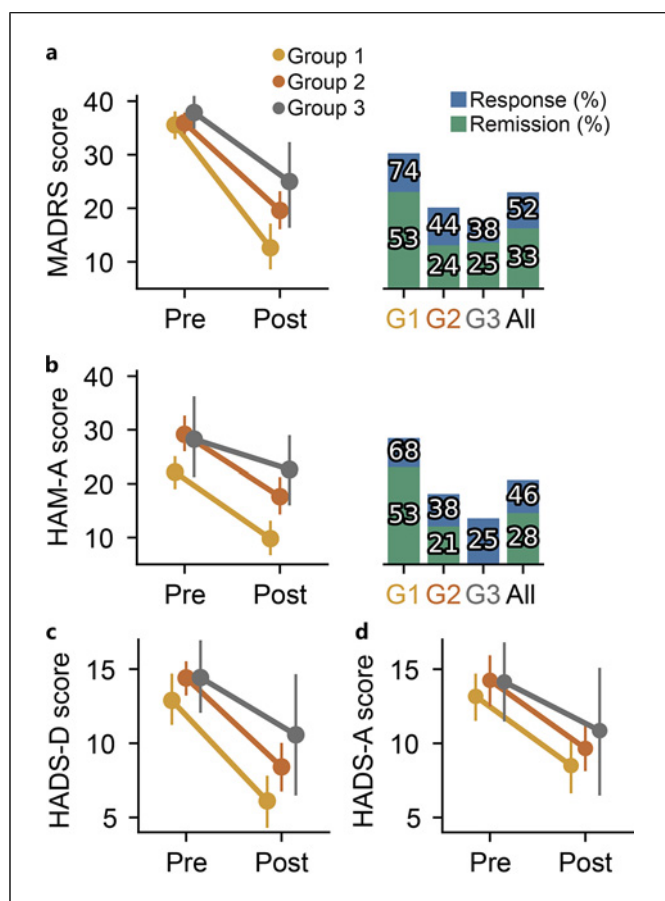
| Demographic measure                                                   | G1:<br>RCT-acceptable     | G2: complex<br>comorbidity | G3: bipolar or<br>neurological disorder | Total                  |
|-----------------------------------------------------------------------|---------------------------|----------------------------|-----------------------------------------|------------------------|
| <i>N</i>                                                              | 19                        | 34                         | 8                                       | 61                     |
| Age, years<br>Median (IQR)                                            | 53 (36–58)                | 33.5 (26–47.8)             | 53 (39.5–62.3)                          | 38 (27–57)             |
| Gender<br>Female                                                      | 8 (42.1%)                 | 16 (47.1%)                 | 3 (37.5%)                               | 27 (44.3%)             |
| Male                                                                  | 11 (57.9%)                | 18 (52.9%)                 | 5 (62.5%)                               | 34 (55.7%)             |
| Handedness<br>Right                                                   | 18 (94.7%)                | 32 (94.1%)                 | 7 (87.5%)                               | 57 (93.4%)             |
| Left                                                                  | 1 (5.3%)                  | 2 (5.9%)                   | 1 (12.5%)                               | 4 (6.6%)               |
| Years of education <sup>a</sup><br>Median (IQR)                       | 12 (12–16)                | 14.5 (12–16.8)             | 16 (14.5–18)                            | 14 (12–16.5)           |
| Age of diagnoses, years <sup>b</sup><br>Median (IQR)                  | 33 (20.5–41)              | 21 (16.3–32)               | 29 (18–40)                              | 24 (17.8–37.3)         |
| Age of symptom onset, years <sup>b</sup><br>Median (IQR)              | 30 (17–37.5)              | 17 (15–24)                 | 21 (17–25.5)                            | 19 (16–33.3)           |
| Diagnostic type<br>Primary MDD                                        | 19 (100%)                 | 0                          | 0                                       | 19 (31.1%)             |
| MDD with complex<br>comorbidity                                       | 0                         | 34 (100%)                  | 0                                       | 34 (55.7%)             |
| Bipolar                                                               | 0                         | 0                          | 7 (87.5%)                               | 7 (11.5%)              |
| Structural brain disease                                              | 0                         | 0                          | 1 (12.5%)                               | 1 (1.6%)               |
| Previous rTMS<br>Yes                                                  | 0                         | 3 (8.8%)                   | 2 (25.0%)                               | 5 (8.2%)               |
| No                                                                    | 19 (100%)                 | 31 (91.2%)                 | 6 (75.0%)                               | 56 (91.8%)             |
| Previous ECT<br>Yes                                                   | 2 (10.5%)                 | 2 (5.9%)                   | 2 (25.0%)                               | 6 (9.8%)               |
| No                                                                    | 17 (89.5%)                | 32 (94.1%)                 | 6 (75.0%)                               | 55 (90.2%)             |
| Treatment regime<br>rTMS sessions, <i>n</i><br>Median (IQR) [min–max] | 20 (20–30) [19–30]        | 27 (20–30) [20–32]         | 20 (20–28.5) [20–30]                    | 21 (20–30) [19–32]     |
| 20 sessions                                                           | 9 (47.4%)                 | 15 (44.1%)                 | 5 (62.5%)                               | 29 (47.5%)             |
| 30 sessions                                                           | 6 (31.6%)                 | 16 (47.1%)                 | 2 (25.0%)                               | 24 (39.3%)             |
| Days from pre-time point to post-time point<br>Median (IQR) [min–max] | 54 (47.5–66.5)<br>[37–78] | 55.5 (49–66.3)<br>[38–108] | 56.5 (50.8–66.8)<br>[37–129]            | 55 (49–67)<br>[37–129] |

<sup>a</sup>*N* = 59, with *n* = 17 for G1: RCT-acceptable. <sup>b</sup>*N* = 60, with *n* = 7 for G3: bipolar or neurological disorder.

(Axilum Robotics Cobot) and a MagPro R30 machine with an actively cooled figure-of-eight coil (Cool-B65). Participants received approximately 20- or 30-weekday sessions of intermittent theta burst stimulation (600 pulses [10], 120% resting motor threshold, online suppl. Material) [11]. Brain stimulation was administered while participants were at rest. Participants were offered the possibility of extending the treatment beyond twenty sessions based on clinical (partial response based on

clinical scales), financial, and pragmatic (ability of the participant to attend the clinic for additional sessions) considerations. Most participants who missed scheduled sessions received the rTMS sessions over a longer period, although a small number deviated from the 20- to 30-weekday plan due to various circumstances (e.g., unrelated illness).

The primary clinical outcome was change in depression severity measured with the clinician-rated



**Fig. 1.** Depression and anxiety outcomes from personalized rTMS treatment. **a** Change in depression severity (MADRS). Mean MADRS scores pre- and post-treatment colored by participant groups are shown on the left. Error bars indicate between-group 95% confidence intervals. The percentage of participants who achieved response (50% reduction in score) and remission ( $\leq 10$  raw score) for each tier, as well as the overall response (“All”), is displayed on the right. **b** Change in anxiety severity (HAM-A). Mean HAM-A scores pre- and post-treatment colored by group are shown on the left. The percentage of participants who achieved response (50% reduction in score) and remission ( $\leq 7$  raw score) for each tier, as well as the overall response (“All”), is displayed on the right. **c** Change in self-reported depression (HADS-D). **d** Change in self-reported anxiety scores (HADS-A, bottom). MADRS, Montgomery-Åsberg Depression Rating Scale; HAM-A, Hamilton Anxiety Rating Scale; HADS-A, Hospital Anxiety and Depression Scale-Anxiety; HADS-D, Hospital Anxiety and Depression Scale-Depression.

Montgomery-Åsberg Depression Rating Scale (MADRS). The secondary outcome was change in anxiety severity measured with the Hamilton Anxiety Rating Scale (HAM-A). Response was defined as a 50% reduction in symptoms. Remission was defined as a score of  $\leq 10$  for the MADRS [12] and  $\leq 7$  for the HAM-A [13]. The MADRS and HAM-

A scales were administered by a psychiatrist (B.B. and J.N.) during the initial consultations and the final review, which took place about 2 weeks after treatment completion. Furthermore, the Hospital Anxiety and Depression Scale (HADS) [14] was used to monitor self-reported participant responses. Note that the HAM-A was not intended to assess anxiety specific to any comorbid condition, including anxiety disorders. It was intended to measure anxious distress in individuals experiencing an acute major depressive episode.

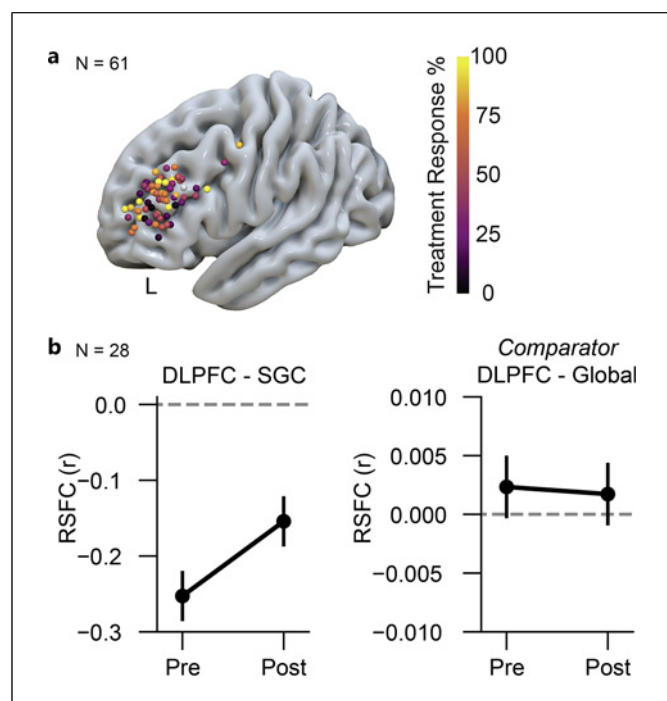
## Results

### MADRS

On the MADRS (Fig. 1a), 52.5% of participants showed a response compared to baseline and 32.8% of the participants achieved remission ( $M_{\text{pre}} = 36.1$ ,  $SD_{\text{pre}} = 5.2$ ,  $M_{\text{post}} = 18.1$ ,  $SD_{\text{post}} = 11.0$ ,  $M_{\text{diff}} = 18.0$ ,  $t(60) = 13.93$ ,  $d = 1.78$ ,  $p < 0.001$ ). Response and remission rates within each of the apriori defined groups [9] were as follows: G1 73.7% response, 52.6% remission ( $M_{\text{pre}} = 35.6$ ,  $SD_{\text{pre}} = 5.2$ ,  $M_{\text{post}} = 12.6$ ,  $SD_{\text{post}} = 9.5$ ); G2 44.1% response, 23.5% remission ( $M_{\text{pre}} = 35.9$ ,  $SD_{\text{pre}} = 5.4$ ,  $M_{\text{post}} = 19.6$ ,  $SD_{\text{post}} = 10.3$ ); G3 37.5% response, 25.0% remission rate ( $M_{\text{pre}} = 37.9$ ,  $SD_{\text{pre}} = 4.4$ ,  $M_{\text{post}} = 25.0$ ,  $SD_{\text{post}} = 12.1$ ). Mixed ANOVA demonstrated a main effect of time ( $F(1, 58) = 214.64$ ,  $\eta^2 = 0.79$ ,  $p < 0.001$ ) and group ( $F(2, 58) = 3.81$ ,  $\eta^2 = 0.12$ ,  $p = 0.028$ ), and an interaction between time and group ( $F(2, 58) = 4.19$ ,  $\eta^2 = 0.13$ ,  $p = 0.020$ ). Follow-up paired  $t$  tests revealed a significant effect of time within each of the groups (G1:  $M_{\text{diff}} = 22.9$ ,  $t(18) = 11.0$ ,  $d = 2.52$ ,  $p < 0.001$ , G2:  $M_{\text{diff}} = 16.4$ ,  $t(33) = 10.1$ ,  $d = 1.73$ ,  $p < 0.001$ , G3:  $M_{\text{diff}} = 12.9$ ,  $t(7) = 3.25$ ,  $d = 1.15$ ,  $p = 0.014$ ,  $\alpha_{\text{Bonferroni}} = 0.0167$ ).

### HAM-A

On the HAM-A (Fig. 1b), 45.9% of participants showed a response compared to baseline and 27.9% of the participants achieved remission ( $M_{\text{pre}} = 26.9$ ,  $SD_{\text{pre}} = 9.5$ ,  $M_{\text{post}} = 15.8$ ,  $SD_{\text{post}} = 9.9$ ,  $M_{\text{diff}} = 11.0$ ,  $t(60) = 9.9$ ,  $d = 1.27$ ,  $p < 0.001$ ). Response and remission rates within each of the groups [9] were as follows: G1 68.4% response, 52.6% remission ( $M_{\text{pre}} = 22.2$ ,  $SD_{\text{pre}} = 6.6$ ,  $M_{\text{post}} = 9.8$ ,  $SD_{\text{post}} = 7.0$ ); G2 38.2% response, 20.6% remission ( $M_{\text{pre}} = 29.1$ ,  $SD_{\text{pre}} = 9.8$ ,  $M_{\text{post}} = 17.6$ ,  $SD_{\text{post}} = 9.9$ ); G3 25.0% response, 0.0% remission ( $M_{\text{pre}} = 28.2$ ,  $SD_{\text{pre}} = 11.0$ ,  $M_{\text{post}} = 22.6$ ,  $SD_{\text{post}} = 9.4$ ). Mixed ANOVA showed main effects of time ( $F(1, 58) = 100.49$ ,  $\eta^2 = 0.63$ ,  $p < 0.001$ ) and group ( $F(2, 58) = 6.46$ ,  $\eta^2 = 0.18$ ,  $p = 0.003$ ), but no interaction between these two factors ( $F(2, 58) = 1.87$ ,  $\eta^2 = 0.06$ ,



**Fig. 2.** Personalized stimulation sites and change in resting-state functional connectivity as a function of personalized rTMS treatment. **a** Left (L) dorsolateral prefrontal cortex stimulation sites colored by the reduction in depressive symptoms (percentage MADRS score) from baseline assessment ( $N = 61$ ). **b** Change in resting-state functional connectivity (RSFC) pre- and post-treatment with repeated measures 95% confidence intervals ( $N = 28$ ). The leftmost subplot shows the change in RSFC between the personalized stimulation site (DLPFC) and the subgenual cortex (SGC), the deep cortical structure on which the fMRI personalization for MDD is based. The rightmost subplot shows the average change in RSFC between the personalized stimulation site (DLPFC) and all other brain regions (global).

$p = 0.16$ ). Paired  $t$  tests revealed a significant effect of time within G1 ( $M_{\text{diff}} = 12.4$ ,  $t(18) = 6.64$ ,  $d = 1.52$ ,  $p < 0.001$ ) and G2 ( $M_{\text{diff}} = 11.5$ ,  $t(33) = 8.80$ ,  $d = 1.51$ ,  $p < 0.001$ ), but not G3 ( $M_{\text{diff}} = 5.6$ ,  $t(7) = 1.24$ ,  $d = 0.44$ ,  $p = 0.26$ ,  $\alpha_{\text{Bonferroni}} = 0.0167$ ).

### HADS

Mixed ANOVA of self-report depression (HADS-D) and anxiety scales (HADS-A) revealed a main effect of time ( $F(1, 54) = 72.29$ ,  $\eta p^2 = 0.57$ ,  $p < 0.001$ ,  $F(1, 54) = 66.93$ ,  $\eta p^2 = 0.55$ ,  $p < 0.001$ , respectively) (Fig. 1c, d). However, no effect of group or group by time interactions was detected, with the exception of a group effect for HADS-D ( $F(2, 54) = 3.21$ ,  $\eta p^2 = 0.11$ ,  $p = 0.048$ ). Note that participants with any missing data were excluded from these analyses ( $N = 57$ ).

### Neuroimaging Outcomes

Personalized targets for the entire cohort are shown in Figure 2a. A subset of participants ( $N = 28$ ) agreed to repeat the neuroimaging session after treatment completion (Supplementary Material). Individual RSFC values were generated by correlating time series from the right SGC (MNI:  $x = 6$ ,  $y = 16$ ,  $z = -10$ ) time series with the individualized left DLPFC rTMS stimulation site time series for both pre- and post-treatment MRI sessions. Time series were derived from a 10-mm sphere centered on each coordinate. As a control analysis, the same left DLPFC time series was correlated with all non-DLPFC voxels. The resulting correlations were then averaged to generate a DLPFC-global connectivity measure.

RSFC in the targeted DLPFC-SGC pathway increased from pre- to post-treatment (Fig. 2b, two-tailed paired  $t$  test:  $M_{\text{diff}} = -0.10$ ,  $t(26) = 4.34$ ,  $d = 0.63$ ,  $p < 0.001$ ). This effect was not evident in a control analysis (average RSFC from the personalized DLPFC site to all other brain regions, paired  $t$  test:  $M_{\text{diff}} = 0.001$ ,  $t(28) = 0.33$ ,  $d = 0.06$ ,  $p = 0.74$ ). Change in RSFC pre- to post-treatment did not correlate with change in MADRS score ( $r_s = 0.078$ ,  $p = 0.70$ ). As a final, exploratory analysis, we evaluated changes in regional brain activity using the fractional amplitude of low-frequency fluctuations (see online suppl. SFig. 1).

### Discussion

Our results provide initial prospective evidence for the clinical effectiveness of precision RSFC-personalized rTMS for refractory MDD using a conventional 4–6-week treatment schedule. The observed overall response (52%) and remission (33%) rates demonstrate numerically superior outcomes compared to a recent large open-label study of standard rTMS that employed standardized clinician-rated scales (31%/23%) [8].

As in standard rTMS [15], treatment was most effective in RCT-acceptable participants [9] (G1; 74%/53%) and decreased with comorbidity (G2; 44%/24%) and bipolar depression or depression co-occurring with neurological conditions (G3; 38%/25%). These findings suggest that the clinical benefits of personalized rTMS targeting based on DLPFC-SGC RSFC may be more likely to translate in less complex cases of treatment-resistant MDD. Critically, the response and remission rates in RCT-acceptable participants demonstrated numerically superior outcomes compared to a meta-analysis of RCT data (45%/22%) [1]. Notably, prior retrospective studies reporting on the association

between DLPFC-SGC RSFC and rTMS treatment outcomes were primarily conducted in cohorts that met RCT criteria (i.e., individuals with relatively pure MDD and minimal comorbidities) [4]. Supporting our clinical findings, neuroimaging comparisons pre- and post-treatment showed reduced anticorrelated RSFC (negative RSFC approaching zero) between the left DLPFC and the right SGC, suggesting that the personalization approach targeted the intended biological substrate.

Our findings highlight the feasibility of RSFC-personalized rTMS in clinical practice and offer needed clinical and neurophysiological evidence supporting the potential benefits of this approach. Future studies outside of the context of a treatment clinic should consider the benefit of collecting a wider breadth of clinical data capturing symptom dimensions of MDD. Prospective RCTs are required to evaluate the clinical gains of fMRI-personalized rTMS compared to standard scalp-based targeting methodologies.

## Acknowledgments

The authors would like to thank the clients of QNC for generously allowing their data to be used for scientific research.

## Statement of Ethics

Treatment procedures were approved by the Australian TGA (DAP21-0020539). This study protocol was reviewed and approved by the UnitingCare (APN 164) and QIMR Berghofer (P3914) Ethics Committees (Brisbane, Australia). Participants provided written informed consent.

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## Conflict of Interest Statement

L.J.H., R.C., C.R., P.E.M., J.N., S.T.T., S.I., J.M., A.Z., B.B., and L.C. are involved in a clinical neuromodulation center (Queensland Neurostimulation Centre, QNC, as trading for Australia Brain Foundation). Data presented in this work have been collected at QNC. L.J.H., R.C., P.E.M., A.Z., and L.C. are not paid by QNC, while B.B., P.E.M., and J.N. (psychiatrists) and S.T.T., S.I., C.R., J.M. (technicians) received monetary compensation for their work with QNC. The remaining authors declare no competing interests.

## Funding Sources

This work was supported by the Australian NHMRC (GN2001283 and GNT2027597, L.C.). A.Z. and L.J.H. were supported by research fellowships from the NHMRC (APP1118153, APP1194070, respectively). The funder had no role in the design, data collection, data analysis, and reporting of this study.

## Author Contributions

L.J.H., R.C., C.R., P.E.M., J.N., S.T.T., S.I., J.M., A.Z., B.B., and L.C. contributed to QNC, which generated the data for this report. L.J.H., L.W., and N.Y.T. analyzed the data. L.J.H. and L.C. wrote the first draft of the manuscript. All authors edited the manuscript.

## Data Availability Statement

The data that support the findings of this study are not publicly available due to restrictions in the participants consent forms and QIMR Berghofer data protection policies but are available from L.C. upon reasonable request after a data-sharing agreement with QIMR Berghofer has been signed.

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